Aluno: Silas Hoffmann Reis Viana

Questões teóricas:

1. O que é um arquivo fonte?
    a. **um arquivo de texto que contém instruções de linguagem de programação.**
    b. um subdiretório que contém os programas.
    c. um arquivo que contém dados para um programa.
    d. um documento que contém os requisitos para um projeto.

2. O que é um registrador?
    a. **parte do sistema de computador que mantém o controle dos parâmetros do sistema.**
    b. uma parte do processador que possui um padrão de bits.
    c. parte do processador que contém o seu número de série único.
    d. parte do bus de sistema que contém dados.

3. Qual o caracter que, na linguagem assembly do MIPS, inicia um comentário?
    a. **#**
    b. $
    c. / /
    d. *

4. Quantos bits há em cada instrução de máquina MIPS?
    a. 8
    b. 16
    c. **32**
    d. instruções diferentes possuem diferentes comprimentos.

5. O que é o contador de programa?
    a. um registrador que mantém a conta do número de erros durante a execução de um programa.
    b. uma parte do processador que contém o endereço da primeira palavra de dados.
    c. uma variável na montadora que os números das linhas do arquivo de origem.
    d. **parte do processador que contém o endereço da próxima instrução de máquina para ser obtida.**

6. Ao executarmos uma instrução, quanto será adicionado ao contador de programa?

a. 1
b. 2
c. 4
d. 8

7. O que é uma diretiva, tal como a diretiva .text?

a. uma instrução em linguagem assembly que resulta em uma instrução em linguagem de máquina.
b. uma das opções de menu do sistema SPIM.
c. uma instrução em linguagem de máquina que faz com que uma operação sobre os dados ocorra.
d. uma declaração que diz o montador algo sobre o que o programador quer, mas não
e. corresponde diretamente a uma instrução de máquina.

8. O que é um endereço simbólico?

a. um local de memória que contém dados simbólicos.
b. um byte na memória que contém o endereço de dados.
c. símbolo dado como argumento para uma diretiva.
d. um nome usado no código-fonte em linguagem assembly para um local na memória.

9. Em qual endereço o simulador MARS coloca a primeira instrução de máquina quando ele está sendo executado?

a. 0x00000000
b. 0x00400000
c. 0x10000000
d. 0xFFFFFFFF

10. Algumas instruções de máquina possuem uma constante como um dos operandos. Como é chamado tal operando?

a. operando imediato
b. operando embutido
c. operando binário
d. operando de máquina

11. Como é chamada uma operação lógica executada entre bits de cada coluna dos operandos para produzir um bit de resultado para cada coluna?

a. operação lógica
b. operação bitwise
c. operação binária
d. operação coluna

12. Quando uma operação é de fato executada, como estão os operandos na ALU?

a. Pelo menos um operando deve ser de 32 bit.
b. Cada operando pode ser de qualquer tamanho.
c. Ambos operandos devem que vir de registros.
d. Cada um dos registradores deve possuir 32 bit.

13. Dezesseis bits de dados de uma instrução de ori são usados como um operando imediato. Durante execução, o que deve ser feito primeiro?

a. Os dados são estendidos em zero à direita por 16 bits.
b. Os dados são estendidos em zero à esquerda por 16 bits.
c. Nada precisa ser feito.
d. Apenas 16 bits são usados pelo outro operando.

14. Qual das instruções seguintes armazenam no registrador $5 um padrão de bits que representa positivo 48?

a. ori $5, $0, 0x48
b. ori $5, $5, 0x48
c. ori $5, $0, 48
d. ori $0, $5, 0x48

15. A instrução de ori pode armazenar o complemento de dois de um número em um registrador?

a. Não.
b. Sim.

16. Qual das instruções seguintes limpa todos os bits no registrador $8 com exceção do byte de baixa ordem que fica inalterado?

a. ori $8, $8, 0xFF
b. ori $8, $0, 0x00FF
c. xori $8, $8, 0xFF
d. andi $8, $8, 0xFF

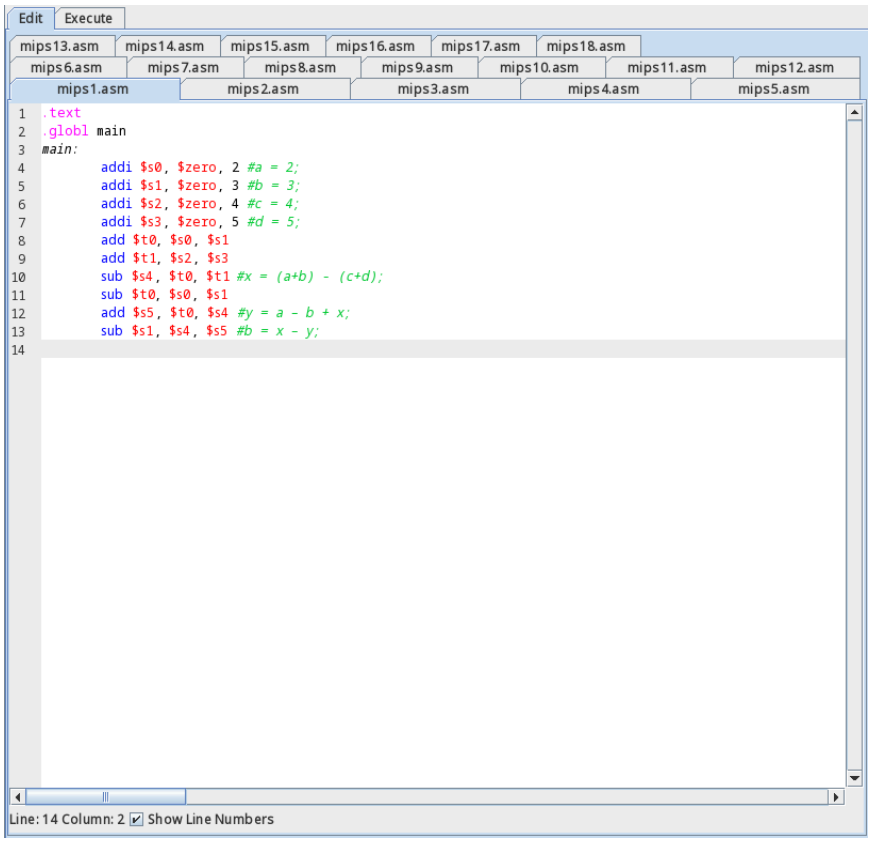
17. Qual é o resultado de um ou exclusivo de padrão sobre ele mesmo?

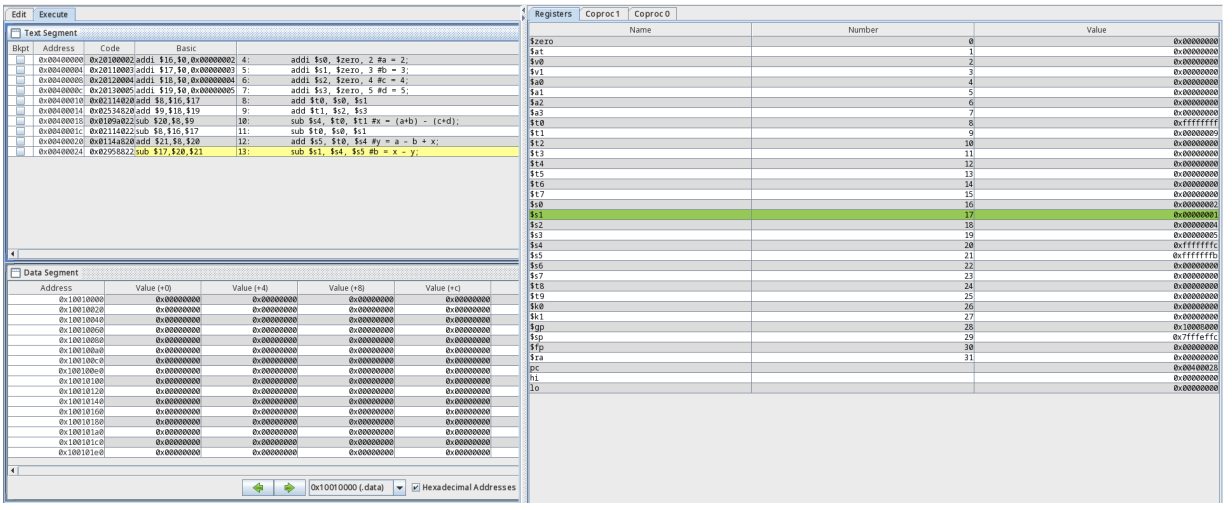
a. Todos os bits em zero.
b. Todos os bits em um.
c. O padrão original utilizado.
d. O resultado é o contrário do original.

18. Todas as instruções de máquina têm os mesmos campos?

a. Não. Diferentes de instruções de máquina possuem campos diferentes.
b. Não. Cada instrução de máquina é completamente diferente de qualquer outra.
c. Sim. Todas as instruções de máquina têm os mesmos campos na mesma ordem.
d. Sim. Todas as instruções de máquina têm os mesmos campos, mas eles podem estar em ordens diferentes.

Exercício 1

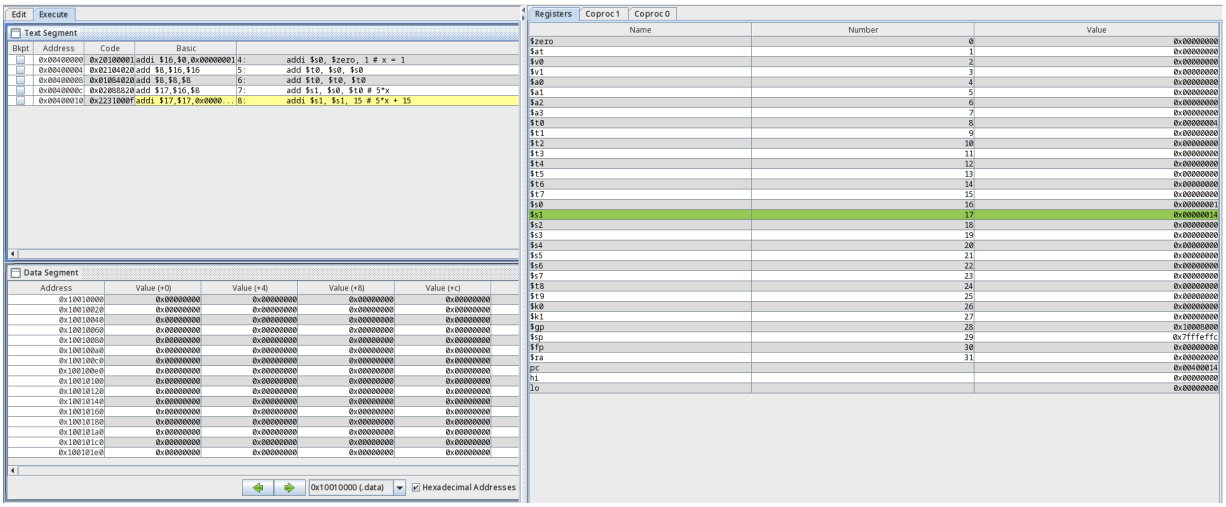

Edit
Execute
mips13.asm
mips14.asm
mips15.asm
mips16.asm
mips17.asm
mips18.asm
mips6.asm
mips7.asm
mips8.asm
mips9.asm
mips10.asm
mips11.asm
mips12.asm
mips1.asm
mips2.asm
mips3.asm
mips4.asm
mips5.asm
1 .text
2 .globl main
3 main:
4 addi $s0, $zero, 2 #a = 2;
5 addi $s1, $zero, 3 #b = 3;
6 addi $s2, $zero, 4 #c = 4;
7 addi $s3, $zero, 5 #d = 5;
8 add $t0, $s0, $s1
9 add $t1, $s2, $s3
10 sub $s4, $t0, $t1 #x = (a+b) - (c+d);
11 sub $t0, $s0, $s1
12 add $s5, $t0, $s4 #y = a - b + x;
13 sub $s1, $s4, $s5 #b = x - y;
14
Line: 14 Column: 2
Show Line Numbers


Edit
Execute
Text Segment
Bkpt Address Code Basic
0x00400000 0x20100002 addi $16,$0,0x00000002 4: addi $s0, $zero, 2 #a = 2;
0x00400004 0x20110003 addi $17,$0,0x00000003 5: addi $s1, $zero, 3 #b = 3;
0x00400008 0x20120004 addi $18,$0,0x00000004 6: addi $s2, $zero, 4 #c = 4;
0x0040000c 0x20130005 addi $19,$0,0x00000005 7: addi $s3, $zero, 5 #d = 5;
0x00400010 0x02114020 add $8,$16,$17 8: add $t0, $s0, $s1
0x00400014 0x02534820 add $9,$18,$19 9: add $t1, $s2, $s3
0x00400018 0x01094822 sub $20,$8,$9 10: sub $s4, $t0, $t1 #x = (a+b) - (c+d);
0x0040001c 0x02114022 sub $8,$16,$17 11: sub $t0, $s0, $s1
0x00400020 0x0114a820 add $21,$8,$20 12: add $s5, $t0, $s4 #y = a - b + x;
0x00400024 0x02958822 sub $17,$20,$21 13: sub $s1, $s4, $s5 #b = x - y;
Data Segment
Address Value (+0) Value (+4) Value (+8) Value (+c)
0x10010000 0x00000000 0x00000000 0x00000000 0x00000000
0x10010020 0x00000000 0x00000000 0x00000000 0x00000000
0x10010040 0x00000000 0x00000000 0x00000000 0x00000000
0x10010060 0x00000000 0x00000000 0x00000000 0x00000000
0x10010080 0x00000000 0x00000000 0x00000000 0x00000000
0x100100a0 0x00000000 0x00000000 0x00000000 0x00000000
0x100100c0 0x00000000 0x00000000 0x00000000 0x00000000
0x100100e0 0x00000000 0x00000000 0x00000000 0x00000000
0x10010100 0x00000000 0x00000000 0x00000000 0x00000000
0x10010120 0x00000000 0x00000000 0x00000000 0x00000000
0x10010140 0x00000000 0x00000000 0x00000000 0x00000000
0x10010160 0x00000000 0x00000000 0x00000000 0x00000000
0x10010180 0x00000000 0x00000000 0x00000000 0x00000000
0x100101a0 0x00000000 0x00000000 0x00000000 0x00000000
0x100101c0 0x00000000 0x00000000 0x00000000 0x00000000
0x100101e0 0x00000000 0x00000000 0x00000000 0x00000000
0x10010000 (.data)
Hexadecimal Addresses
Registers
Coproc 1
Coproc 0
Name Number Value
$zero 0 0x00000000
$at 1 0x00000000
$v0 2 0x00000000
$v1 3 0x00000000
$a0 4 0x00000000
$a1 5 0x00000000
$a2 6 0x00000000
$a3 7 0x00000000
$t0 8 0xffffffff
$t1 9 0x00000005
$t2 10 0x00000000
$t3 11 0x00000000
$t4 12 0x00000000
$t5 13 0x00000000
$t6 14 0x00000000
$t7 15 0x00000000
$s0 16 0x00000002
$s1 17 0x00000001
$s2 18 0x00000004
$s3 19 0x00000005
$s4 20 0xfffffffc
$s5 21 0xfffffffb
$s6 22 0x00000000
$s7 23 0x00000000
$t8 24 0x00000000
$t9 25 0x00000000
$k0 26 0x00000000
$k1 27 0x00000000
$gp 28 0x10008000
$sp 29 0x7fffeffc
$fp 30 0x00000000
$ra 31 0x00000000
pc 0x00400028
hi 0x00000000
lo 0x00000000

Exercício 2

```
.text
.globl main
main:
        addi $s0, $zero, 1 # x = 1
        add $t0, $s0, $s0
        add $t0, $t0, $t0
        add $s1, $s0, $t0 # 5*x
        addi $s1, $s1, 15 # 5*x + 15
```

Line: 9 Column: 2 Show Line Numbers



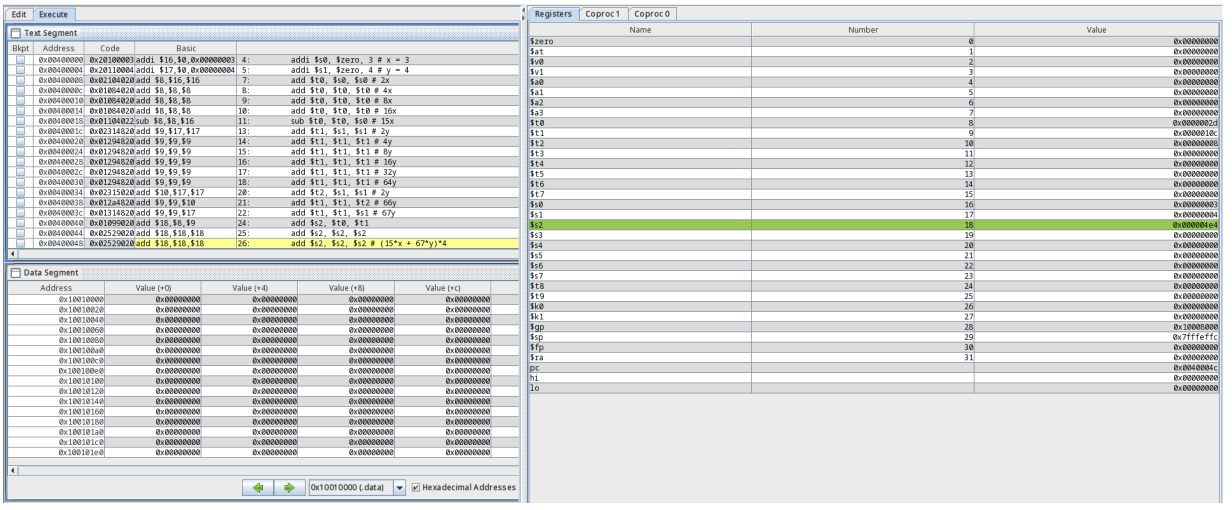
Exercício 3

Edit Execute

mips13.asm mips14.asm mips15.asm mips16.asm mips17.asm mips18.asm
mips6.asm mips7.asm mips8.asm mips9.asm mips10.asm mips11.asm mips12.asm
mips1.asm mips2.asm mips3.asm mips4.asm mips5.asm

```
1  .text
2  .globl main
3  main:
4          addi $s0, $zero, 3 # x = 3
5          addi $s1, $zero, 4 # y = 4
6
7          add $t0, $s0, $s0 # 2x
8          add $t0, $t0, $t0 # 4x
9          add $t0, $t0, $t0 # 8x
10         add $t0, $t0, $t0 # 16x
11         sub $t0, $t0, $s0 # 15x
12
13         add $t1, $s1, $s1 # 2y
14         add $t1, $t1, $t1 # 4y
15         add $t1, $t1, $t1 # 8y
16         add $t1, $t1, $t1 # 16y
17         add $t1, $t1, $t1 # 32y
18         add $t1, $t1, $t1 # 64y
19
20         add $t2, $s1, $s1 # 2y
21         add $t1, $t1, $t2 # 66y
22         add $t1, $t1, $s1 # 67y
23
24         add $s2, $t0, $t1
25         add $s2, $s2, $s2
26         add $s2, $s2, $s2 # (15*x + 67*y)*4
27
```

Line: 27 Column: 2 ☑ Show Line Numbers



Exercício 4

```
.text
.globl main
main:
        addi $s0, $zero, 3 # x = 3
        addi $s1, $zero, 4 # y = 4

        sll $t0, $s0, 4 # t0 = 16x
        sub $t0, $t0, $s0 # t0 = 15x

        sll $t1, $s1, 6 # 64y
        sll $t2, $s1, 1 # 2y
        add $t1, $t1, $t2 # 66y
        add $t1, $t1, $s1 # 67y

        add $s2, $t0, $t1 # 15x + 67y
        sll $s2, $s2, 2 # (15x + 67y)* 4
```

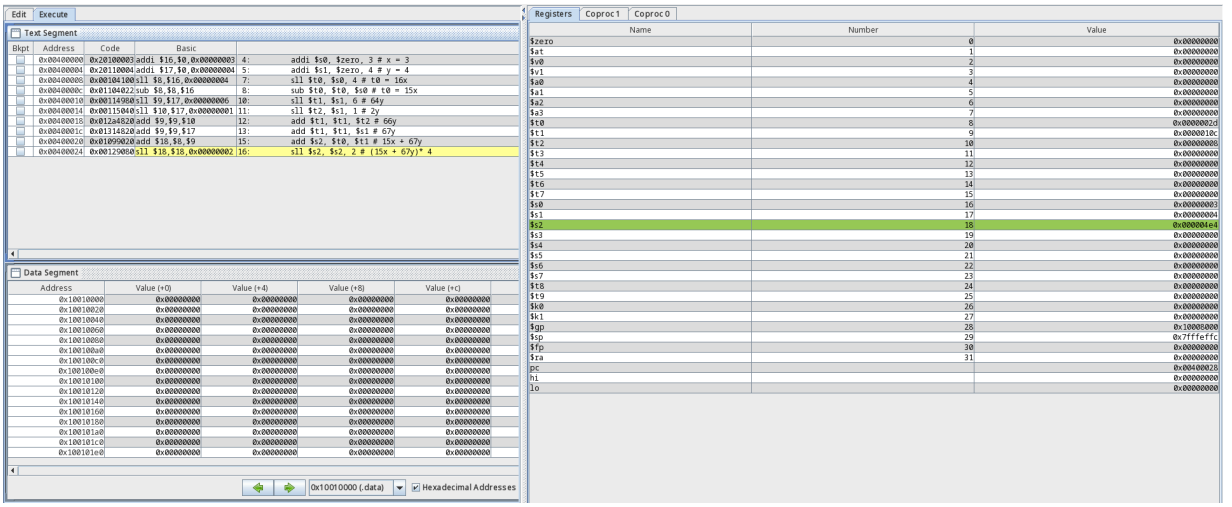
Edit | Execute

mips13.asm | mips14.asm | mips15.asm | mips16.asm | mips17.asm | mips18.asm

mips6.asm | mips7.asm | mips8.asm | mips9.asm | mips10.asm | mips11.asm | mips12.asm

mips1.asm | mips2.asm | mips3.asm | mips4.asm | mips5.asm

Line: 17 Column: 1 ☑ Show Line Numbers



Text Segment

Bkpt	Address	Code	Basic	
	0x00400000	0x20100003	addi $16,$0,0x00000003	4: addi $s0, $zero, 3 # x = 3
	0x00400004	0x20110004	addi $17,$0,0x00000004	5: addi $s1, $zero, 4 # y = 4
	0x00400008	0x00104100	sll $8,$16,0x00000004	7: sll $t0, $s0, 4 # t0 = 16x
	0x0040000c	0x01104022	sub $8,$8,$16	8: sub $t0, $t0, $s0 # t0 = 15x
	0x00400010	0x00114980	sll $9,$17,0x00000006	10: sll $t1, $s1, 6 # 64y
	0x00400014	0x00115040	sll $10,$17,0x00000001	11: sll $t2, $s1, 1 # 2y
	0x00400018	0x012a4820	add $9,$9,$10	12: add $t1, $t1, $t2 # 66y
	0x0040001c	0x01314820	add $9,$9,$17	13: add $t1, $t1, $s1 # 67y
	0x00400020	0x01099020	add $18,$8,$9	15: add $s2, $t0, $t1 # 15x + 67y
	0x00400024	0x00129080	sll $18,$18,0x00000002	16: sll $s2, $s2, 2 # (15x + 67y)* 4

Data Segment

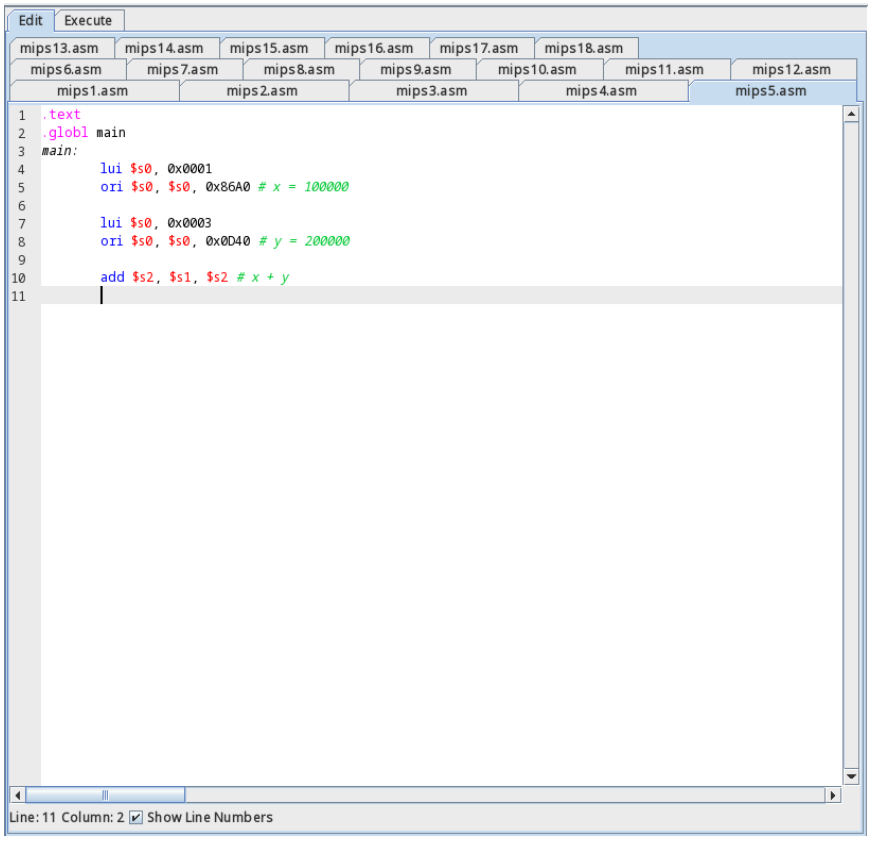
Address	Value (+0)	Value (+4)	Value (+8)	Value (+c)
0x10010000	0x00000000	0x00000000	0x00000000	0x00000000
0x10010020	0x00000000	0x00000000	0x00000000	0x00000000
0x10010040	0x00000000	0x00000000	0x00000000	0x00000000
0x10010060	0x00000000	0x00000000	0x00000000	0x00000000
0x10010080	0x00000000	0x00000000	0x00000000	0x00000000
0x100100a0	0x00000000	0x00000000	0x00000000	0x00000000
0x100100c0	0x00000000	0x00000000	0x00000000	0x00000000
0x100100e0	0x00000000	0x00000000	0x00000000	0x00000000
0x10010100	0x00000000	0x00000000	0x00000000	0x00000000
0x10010120	0x00000000	0x00000000	0x00000000	0x00000000
0x10010140	0x00000000	0x00000000	0x00000000	0x00000000
0x10010160	0x00000000	0x00000000	0x00000000	0x00000000
0x10010180	0x00000000	0x00000000	0x00000000	0x00000000
0x100101a0	0x00000000	0x00000000	0x00000000	0x00000000
0x100101c0	0x00000000	0x00000000	0x00000000	0x00000000
0x100101e0	0x00000000	0x00000000	0x00000000	0x00000000

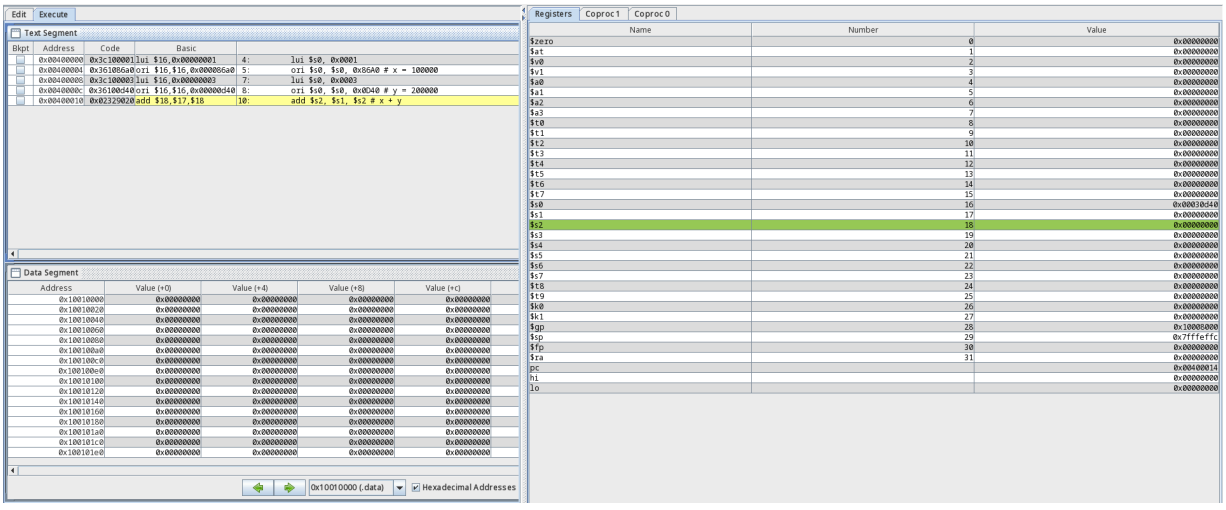
0x10010000 (.data) ☑ Hexadecimal Addresses

Registers | Coproc 1 | Coproc 0

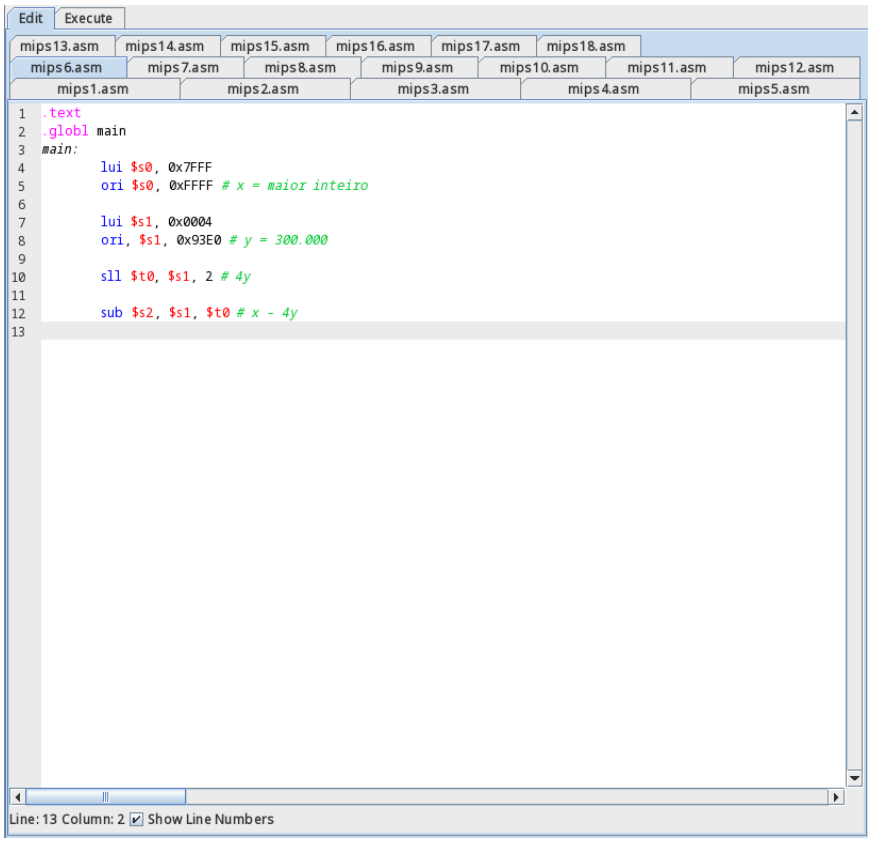
Name	Number	Value
$zero	0	0x00000000
$at	1	0x00000000
$v0	2	0x00000000
$v1	3	0x00000000
$a0	4	0x00000000
$a1	5	0x00000000
$a2	6	0x00000000
$a3	7	0x00000000
$t0	8	0x0000002d
$t1	9	0x0000010c
$t2	10	0x00000000
$t3	11	0x00000000
$t4	12	0x00000000
$t5	13	0x00000000
$t6	14	0x00000000
$t7	15	0x00000000
$s0	16	0x00000003
$s1	17	0x00000004
$s2	18	0x000001e4
$s3	19	0x00000000
$s4	20	0x00000000
$s5	21	0x00000000
$s6	22	0x00000000
$s7	23	0x00000000
$t8	24	0x00000000
$t9	25	0x00000000
$k0	26	0x00000000
$k1	27	0x00000000
$gp	28	0x10008000
$sp	29	0x7fffeffc
$fp	30	0x00000000
$ra	31	0x00000000
pc		0x00400028
hi		0x00000000
lo		0x00000000

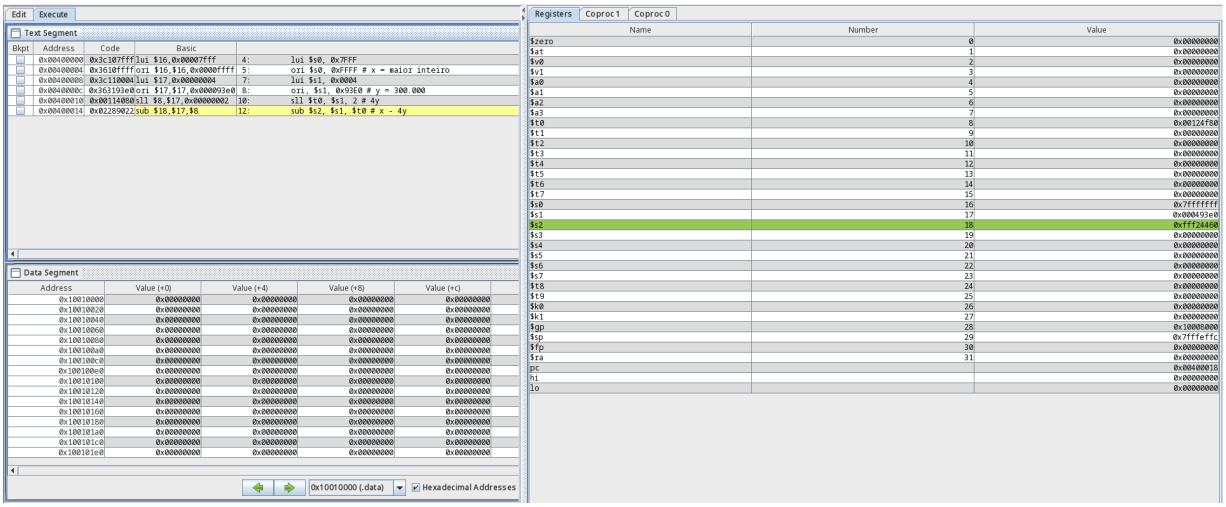
Exercício 5


Edit
Execute
mips13.asm
mips14.asm
mips15.asm
mips16.asm
mips17.asm
mips18.asm
mips6.asm
mips7.asm
mips8.asm
mips9.asm
mips10.asm
mips11.asm
mips12.asm
mips1.asm
mips2.asm
mips3.asm
mips4.asm
mips5.asm
1 .text
2 .globl main
3 main:
4 lui $s0, 0x0001
5 ori $s0, $s0, 0x86A0 # x = 100000
6
7 lui $s0, 0x0003
8 ori $s0, $s0, 0x0D40 # y = 200000
9
10 add $s2, $s1, $s2 # x + y
11
Line: 11 Column: 2
Show Line Numbers

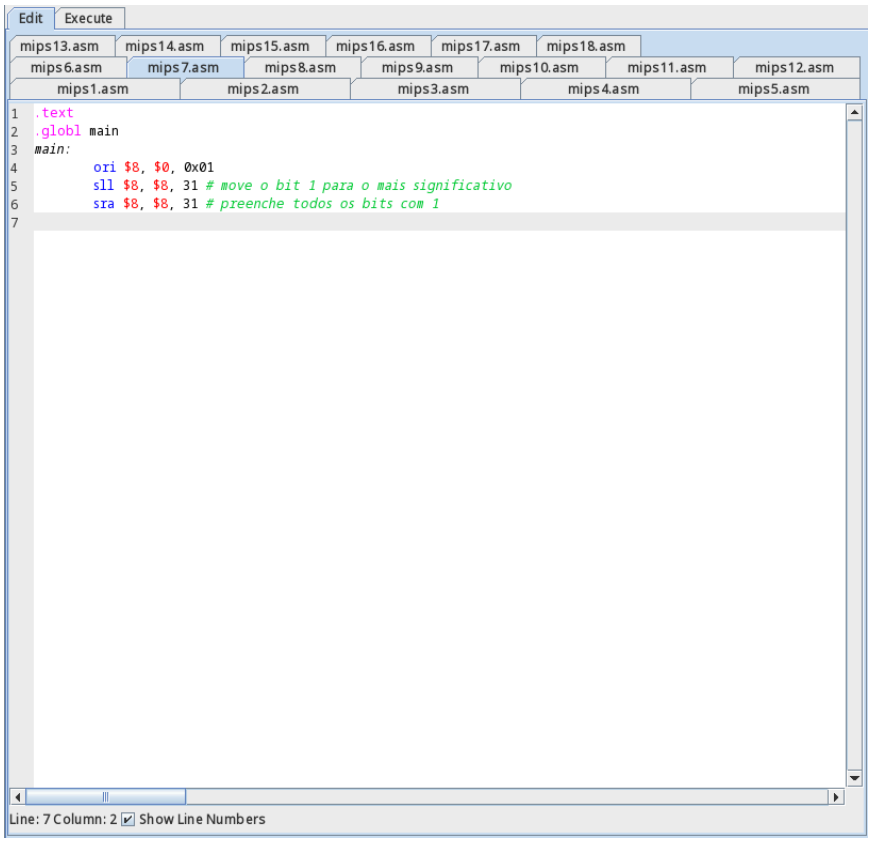

Edit
Execute
Text Segment
Bkpt
Address
Code
Basic
0x00400000 0x3c100001 lui $16,0x00000001 4: lui $s0, 0x0001
0x00400004 0x361086a0 ori $16,$16,0x000086a0 5: ori $s0, $s0, 0x86A0 # x = 100000
0x00400008 0x3c100003 lui $16,0x00000003 7: lui $s0, 0x0003
0x0040000c 0x36100d40 ori $16,$16,0x00000d40 8: ori $s0, $s0, 0x0D40 # y = 200000
0x00400010 0x02329020 add $18,$17,$18 10: add $s2, $s1, $s2 # x + y
Data Segment
Address
Value (+0)
Value (+4)
Value (+8)
Value (+c)
0x10010000
0x10010020
0x10010040
0x10010060
0x10010080
0x100100a0
0x100100c0
0x100100e0
0x10010100
0x10010120
0x10010140
0x10010160
0x10010180
0x100101a0
0x100101c0
0x100101e0
0x00000000
0x10010000 (.data)
Hexadecimal Addresses
Registers
Coproc 1
Coproc 0
Name
Number
Value
$zero 0 0x00000000
$at 1 0x00000000
$v0 2 0x00000000
$v1 3 0x00000000
$a0 4 0x00000000
$a1 5 0x00000000
$a2 6 0x00000000
$a3 7 0x00000000
$t0 8 0x00000000
$t1 9 0x00000000
$t2 10 0x00000000
$t3 11 0x00000000
$t4 12 0x00000000
$t5 13 0x00000000
$t6 14 0x00000000
$t7 15 0x00000000
$s0 16 0x00010d40
$s1 17 0x00000000
$s2 18 0x00000000
$s3 19 0x00000000
$s4 20 0x00000000
$s5 21 0x00000000
$s6 22 0x00000000
$s7 23 0x00000000
$t8 24 0x00000000
$t9 25 0x00000000
$k0 26 0x00000000
$k1 27 0x00000000
$gp 28 0x10008000
$sp 29 0x7fffeffc
$fp 30 0x00000000
$ra 31 0x00000000
pc 0x00400014
hi 0x00000000
lo 0x00000000

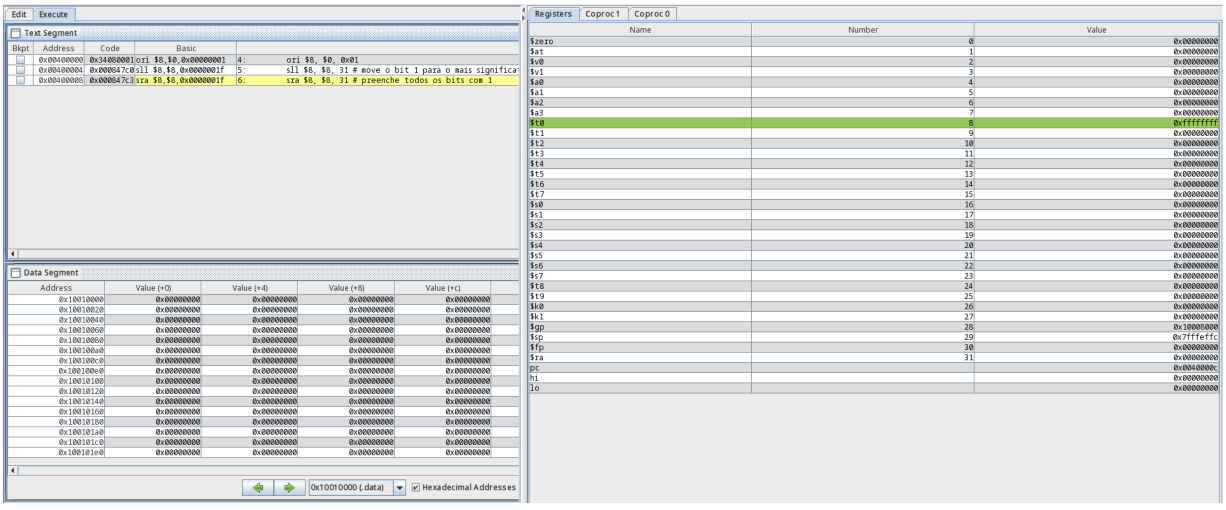
Exercício 6


Edit
Execute
mips13.asm
mips14.asm
mips15.asm
mips16.asm
mips17.asm
mips18.asm
mips6.asm
mips7.asm
mips8.asm
mips9.asm
mips10.asm
mips11.asm
mips12.asm
mips1.asm
mips2.asm
mips3.asm
mips4.asm
mips5.asm
1 .text
2 .globl main
3 main:
4 lui $s0, 0x7FFF
5 ori $s0, 0xFFFF # x = maior inteiro
6
7 lui $s1, 0x0004
8 ori, $s1, 0x93E0 # y = 300.000
9
10 sll $t0, $s1, 2 # 4y
11
12 sub $s2, $s1, $t0 # x - 4y
13
Line: 13 Column: 2
Show Line Numbers

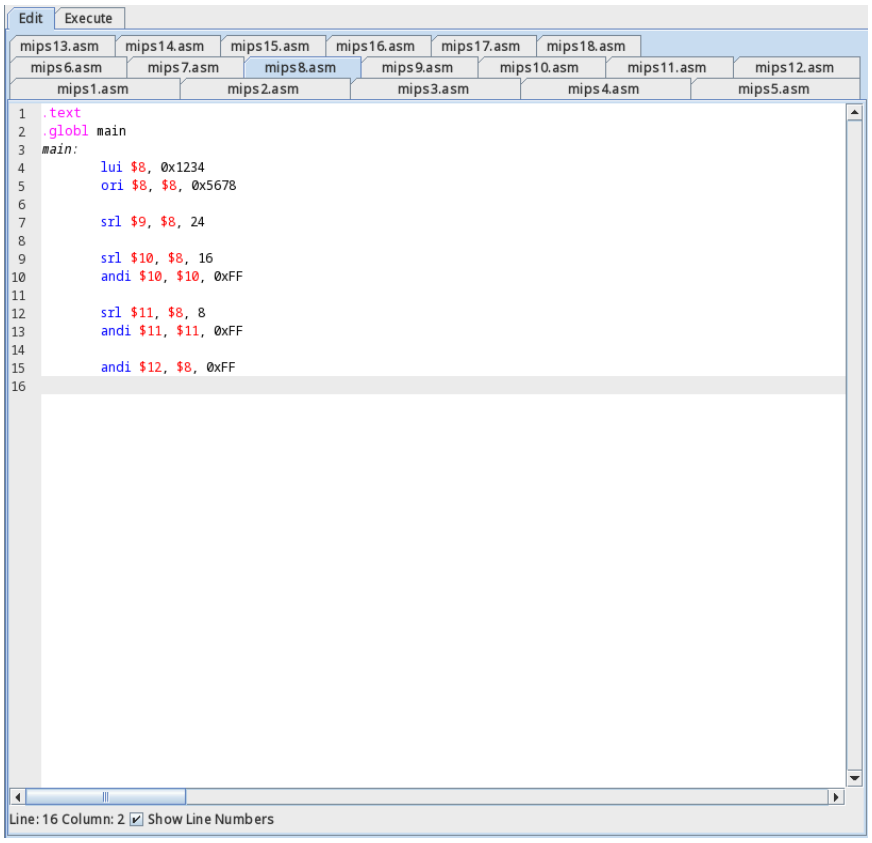

Edit
Execute
Text Segment
Bkpt Address Code Basic
0x00400000 0x3c107fff lui $16,0x00007fff 4: lui $s0, 0x7FFF
0x00400004 0x3610ffff ori $16,$16,0x0000ffff 5: ori $s0, 0xFFFF # x = maior inteiro
0x00400008 0x3c110004 lui $17,0x00000004 7: lui $s1, 0x0004
0x0040000c 0x363193e0 ori $17,$17,0x000093e0 8: ori, $s1, 0x93E0 # y = 300.000
0x00400010 0x00114080 sll $8,$17,0x00000002 10: sll $t0, $s1, 2 # 4y
0x00400014 0x02289022 sub $18,$17,$8 12: sub $s2, $s1, $t0 # x - 4y
Data Segment
Address Value (+0) Value (+4) Value (+8) Value (+c)
0x10010000 0x00000000 0x00000000 0x00000000 0x00000000
0x10010020 0x00000000 0x00000000 0x00000000 0x00000000
0x10010040 0x00000000 0x00000000 0x00000000 0x00000000
0x10010060 0x00000000 0x00000000 0x00000000 0x00000000
0x10010080 0x00000000 0x00000000 0x00000000 0x00000000
0x100100a0 0x00000000 0x00000000 0x00000000 0x00000000
0x100100c0 0x00000000 0x00000000 0x00000000 0x00000000
0x100100e0 0x00000000 0x00000000 0x00000000 0x00000000
0x10010100 0x00000000 0x00000000 0x00000000 0x00000000
0x10010120 0x00000000 0x00000000 0x00000000 0x00000000
0x10010140 0x00000000 0x00000000 0x00000000 0x00000000
0x10010160 0x00000000 0x00000000 0x00000000 0x00000000
0x10010180 0x00000000 0x00000000 0x00000000 0x00000000
0x100101a0 0x00000000 0x00000000 0x00000000 0x00000000
0x100101c0 0x00000000 0x00000000 0x00000000 0x00000000
0x100101e0 0x00000000 0x00000000 0x00000000 0x00000000
0x10010000 (.data)
Hexadecimal Addresses
Registers
Coproc 1
Coproc 0
Name Number Value
$zero 0 0x00000000
$at 1 0x00000000
$v0 2 0x00000000
$v1 3 0x00000000
$a0 4 0x00000000
$a1 5 0x00000000
$a2 6 0x00000000
$a3 7 0x00000000
$t0 8 0x00124f80
$t1 9 0x00000000
$t2 10 0x00000000
$t3 11 0x00000000
$t4 12 0x00000000
$t5 13 0x00000000
$t6 14 0x00000000
$t7 15 0x00000000
$s0 16 0x7fffffff
$s1 17 0x000493e0
$s2 18 0xfff24460
$s3 19 0x00000000
$s4 20 0x00000000
$s5 21 0x00000000
$s6 22 0x00000000
$s7 23 0x00000000
$t8 24 0x00000000
$t9 25 0x00000000
$k0 26 0x00000000
$k1 27 0x00000000
$gp 28 0x10008000
$sp 29 0x7fffeffc
$fp 30 0x00000000
$ra 31 0x00000000
pc 0x00400018
hi 0x00000000
lo 0x00000000

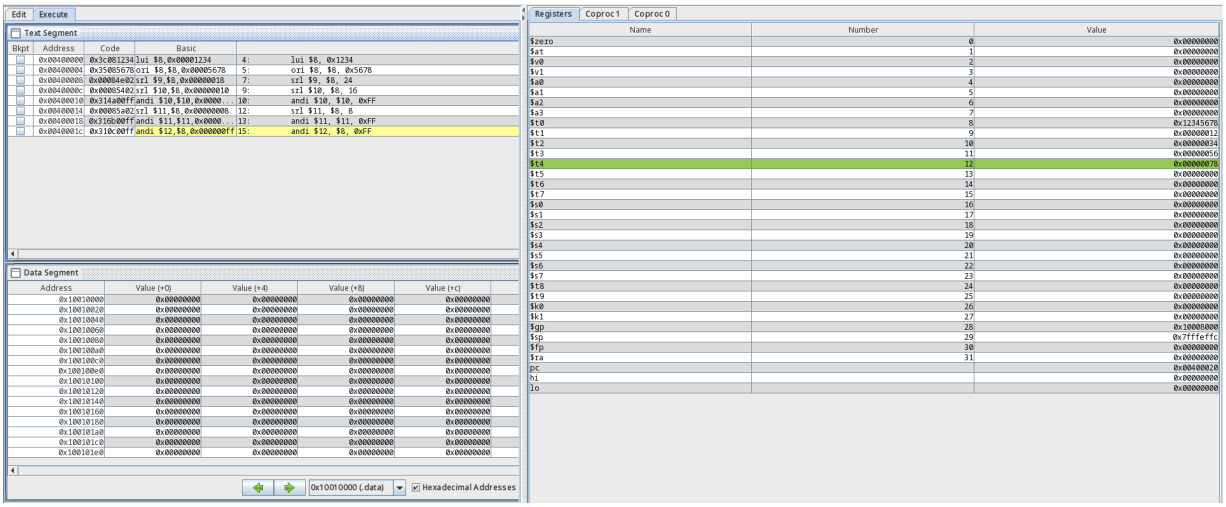
Exercício 7


Edit
Execute
mips13.asm
mips14.asm
mips15.asm
mips16.asm
mips17.asm
mips18.asm
mips6.asm
mips7.asm
mips8.asm
mips9.asm
mips10.asm
mips11.asm
mips12.asm
mips1.asm
mips2.asm
mips3.asm
mips4.asm
mips5.asm
1 .text
2 .globl main
3 main:
4 ori $8, $0, 0x01
5 sll $8, $8, 31 # move o bit 1 para o mais significativo
6 sra $8, $8, 31 # preenche todos os bits com 1
7
Line: 7 Column: 2
Show Line Numbers

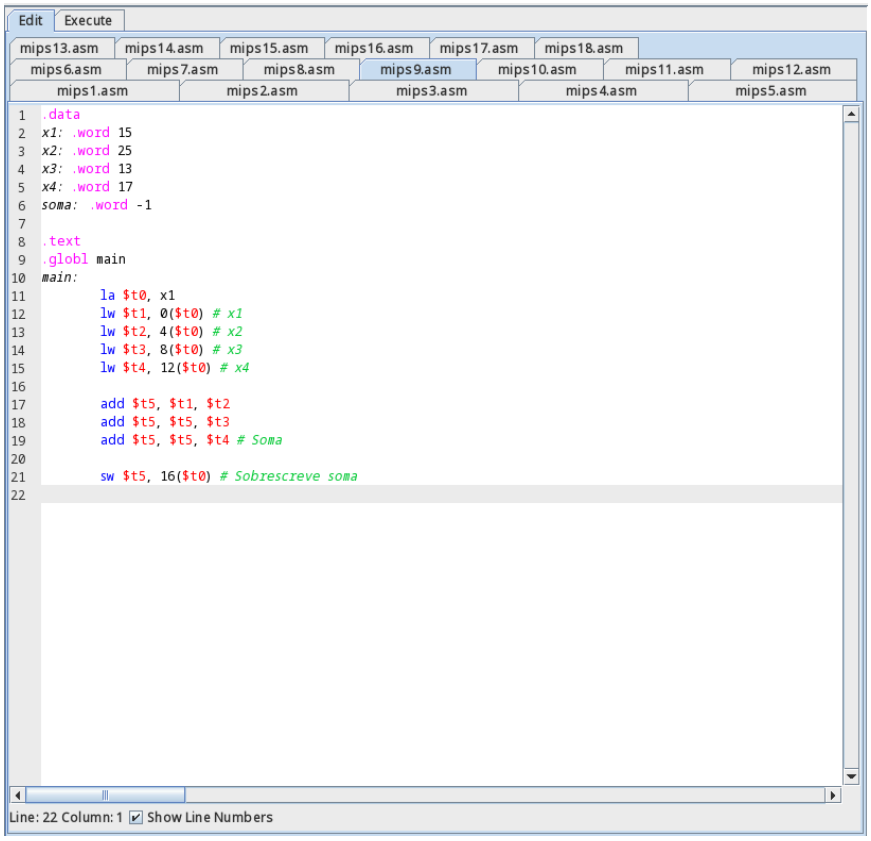

Edit
Execute
Text Segment
Bkpt
Address
Code
Basic
0x00400000 0x34080001 ori $8,$0,0x00000001 4: ori $8, $0, 0x01
0x00400004 0x000847c0 sll $8,$8,0x0000001f 5: sll $8, $8, 31 # move o bit 1 para o mais significativ
0x00400008 0x000847c3 sra $8,$8,0x0000001f 6: sra $8, $8, 31 # preenche todos os bits com 1
Data Segment
Address
Value (+0)
Value (+4)
Value (+8)
Value (+c)
0x10010000 0x00000000 0x00000000 0x00000000 0x00000000
0x10010020 0x00000000 0x00000000 0x00000000 0x00000000
0x10010040 0x00000000 0x00000000 0x00000000 0x00000000
0x10010060 0x00000000 0x00000000 0x00000000 0x00000000
0x10010080 0x00000000 0x00000000 0x00000000 0x00000000
0x100100a0 0x00000000 0x00000000 0x00000000 0x00000000
0x100100c0 0x00000000 0x00000000 0x00000000 0x00000000
0x100100e0 0x00000000 0x00000000 0x00000000 0x00000000
0x10010100 0x00000000 0x00000000 0x00000000 0x00000000
0x10010120 0x00000000 0x00000000 0x00000000 0x00000000
0x10010140 0x00000000 0x00000000 0x00000000 0x00000000
0x10010160 0x00000000 0x00000000 0x00000000 0x00000000
0x10010180 0x00000000 0x00000000 0x00000000 0x00000000
0x100101a0 0x00000000 0x00000000 0x00000000 0x00000000
0x100101c0 0x00000000 0x00000000 0x00000000 0x00000000
0x100101e0 0x00000000 0x00000000 0x00000000 0x00000000
0x10010000 (.data)
Hexadecimal Addresses
Registers
Coproc 1
Coproc 0
Name
Number
Value
$zero 0 0x00000000
$at 1 0x00000000
$v0 2 0x00000000
$v1 3 0x00000000
$a0 4 0x00000000
$a1 5 0x00000000
$a2 6 0x00000000
$a3 7 0x00000000
$t0 8 0xffffffff
$t1 9 0x00000000
$t2 10 0x00000000
$t3 11 0x00000000
$t4 12 0x00000000
$t5 13 0x00000000
$t6 14 0x00000000
$t7 15 0x00000000
$s0 16 0x00000000
$s1 17 0x00000000
$s2 18 0x00000000
$s3 19 0x00000000
$s4 20 0x00000000
$s5 21 0x00000000
$s6 22 0x00000000
$s7 23 0x00000000
$t8 24 0x00000000
$t9 25 0x00000000
$k0 26 0x00000000
$k1 27 0x00000000
$gp 28 0x10008000
$sp 29 0x7fffeffc
$fp 30 0x00000000
$ra 31 0x00000000
pc 0x0040000c
hi 0x00000000
lo 0x00000000

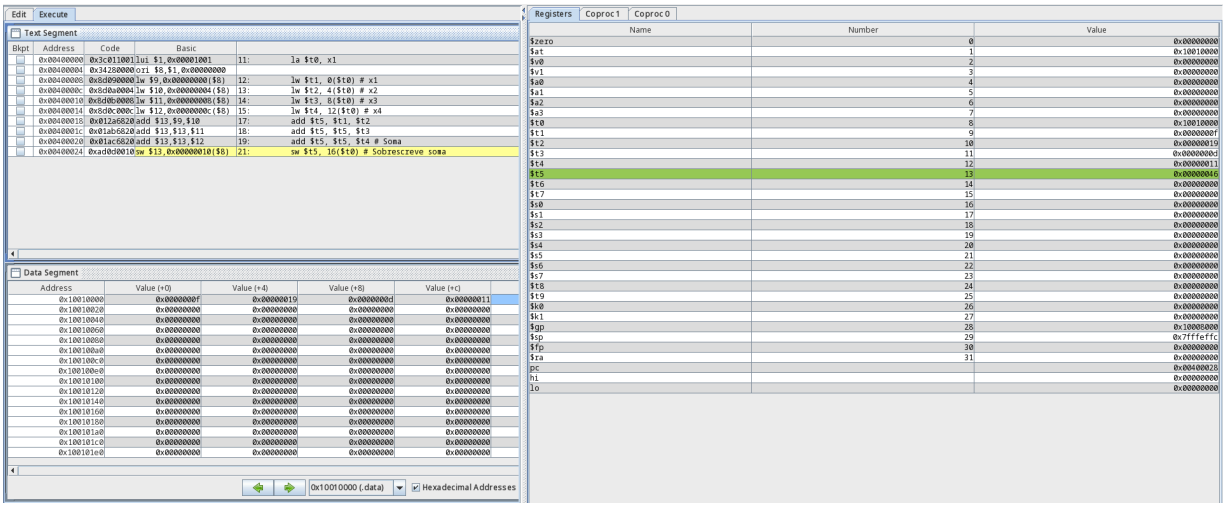
Exercício 8


Edit
Execute
mips13.asm
mips14.asm
mips15.asm
mips16.asm
mips17.asm
mips18.asm
mips6.asm
mips7.asm
mips8.asm
mips9.asm
mips10.asm
mips11.asm
mips12.asm
mips1.asm
mips2.asm
mips3.asm
mips4.asm
mips5.asm
1 .text
2 .globl main
3 main:
4 lui $8, 0x1234
5 ori $8, $8, 0x5678
6
7 srl $9, $8, 24
8
9 srl $10, $8, 16
10 andi $10, $10, 0xFF
11
12 srl $11, $8, 8
13 andi $11, $11, 0xFF
14
15 andi $12, $8, 0xFF
16
Line: 16 Column: 2
Show Line Numbers

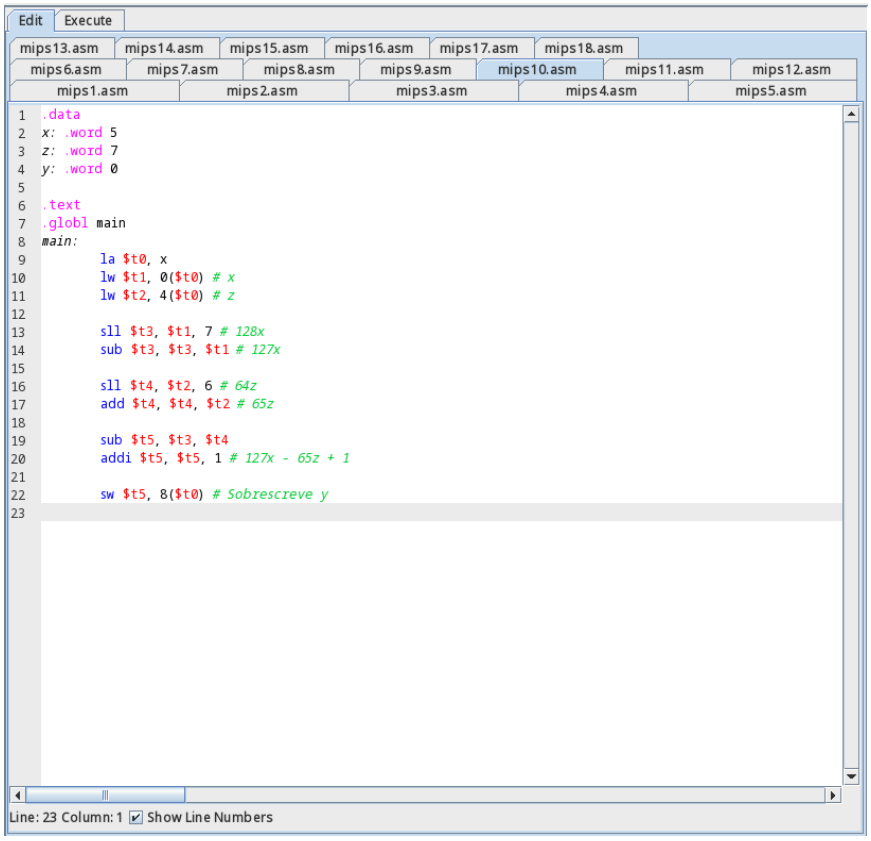

Edit
Execute
Text Segment
Bkpt
Address
Code
Basic
0x00400000 0x3c081234 lui $8,0x00001234 4: lui $8, 0x1234
0x00400004 0x35085678 ori $8,$8,0x00005678 5: ori $8, $8, 0x5678
0x00400008 0x00094e02 srl $9,$8,0x00000018 7: srl $9, $8, 24
0x0040000c 0x00085402 srl $10,$8,0x00000010 9: srl $10, $8, 16
0x00400010 0x314a00ff andi $10,$10,0x0000... 10: andi $10, $10, 0xFF
0x00400014 0x00085a02 srl $11,$8,0x00000008 12: srl $11, $8, 8
0x00400018 0x316b00ff andi $11,$11,0x0000... 13: andi $11, $11, 0xFF
0x0040001c 0x310c00ff andi $12,$8,0x000000ff 15: andi $12, $8, 0xFF
Data Segment
Address
Value (+0)
Value (+4)
Value (+8)
Value (+c)
0x10010000 0x00000000 0x00000000 0x00000000 0x00000000
0x10010020 0x00000000 0x00000000 0x00000000 0x00000000
0x10010040 0x00000000 0x00000000 0x00000000 0x00000000
0x10010060 0x00000000 0x00000000 0x00000000 0x00000000
0x10010080 0x00000000 0x00000000 0x00000000 0x00000000
0x100100a0 0x00000000 0x00000000 0x00000000 0x00000000
0x100100c0 0x00000000 0x00000000 0x00000000 0x00000000
0x100100e0 0x00000000 0x00000000 0x00000000 0x00000000
0x10010100 0x00000000 0x00000000 0x00000000 0x00000000
0x10010120 0x00000000 0x00000000 0x00000000 0x00000000
0x10010140 0x00000000 0x00000000 0x00000000 0x00000000
0x10010160 0x00000000 0x00000000 0x00000000 0x00000000
0x10010180 0x00000000 0x00000000 0x00000000 0x00000000
0x100101a0 0x00000000 0x00000000 0x00000000 0x00000000
0x100101c0 0x00000000 0x00000000 0x00000000 0x00000000
0x100101e0 0x00000000 0x00000000 0x00000000 0x00000000
0x10010000 (.data)
Hexadecimal Addresses
Registers
Coproc 1
Coproc 0
Name
Number
Value
$zero 0 0x00000000
$at 1 0x00000000
$v0 2 0x00000000
$v1 3 0x00000000
$a0 4 0x00000000
$a1 5 0x00000000
$a2 6 0x00000000
$a3 7 0x00000000
$t0 8 0x12345678
$t1 9 0x00000012
$t2 10 0x00000034
$t3 11 0x00000056
$t4 12 0x00000078
$t5 13 0x00000000
$t6 14 0x00000000
$t7 15 0x00000000
$s0 16 0x00000000
$s1 17 0x00000000
$s2 18 0x00000000
$s3 19 0x00000000
$s4 20 0x00000000
$s5 21 0x00000000
$s6 22 0x00000000
$s7 23 0x00000000
$t8 24 0x00000000
$t9 25 0x00000000
$k0 26 0x00000000
$k1 27 0x00000000
$gp 28 0x10008000
$sp 29 0x7fffeffc
$fp 30 0x00000000
$ra 31 0x00000000
pc 0x00400020
hi 0x00000000
lo 0x00000000

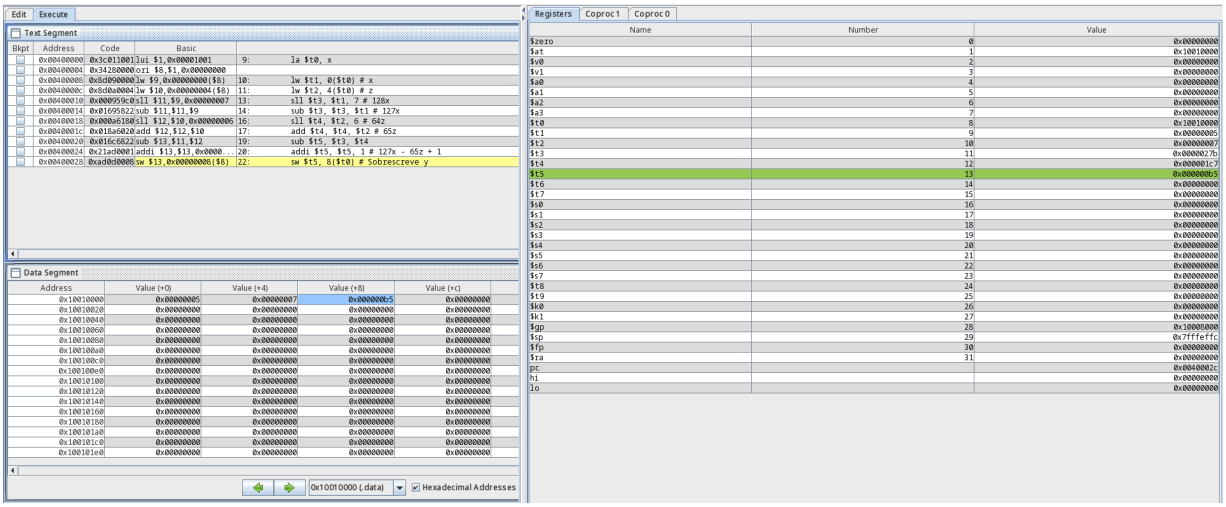
Exercício 9


Edit
Execute
mips13.asm
mips14.asm
mips15.asm
mips16.asm
mips17.asm
mips18.asm
mips6.asm
mips7.asm
mips8.asm
mips9.asm
mips10.asm
mips11.asm
mips12.asm
mips1.asm
mips2.asm
mips3.asm
mips4.asm
mips5.asm
1 .data
2 x1: .word 15
3 x2: .word 25
4 x3: .word 13
5 x4: .word 17
6 soma: .word -1
7
8 .text
9 .globl main
10 main:
11 la $t0, x1
12 lw $t1, 0($t0) # x1
13 lw $t2, 4($t0) # x2
14 lw $t3, 8($t0) # x3
15 lw $t4, 12($t0) # x4
16
17 add $t5, $t1, $t2
18 add $t5, $t5, $t3
19 add $t5, $t5, $t4 # Soma
20
21 sw $t5, 16($t0) # Sobrescreve soma
22
Line: 22 Column: 1
Show Line Numbers

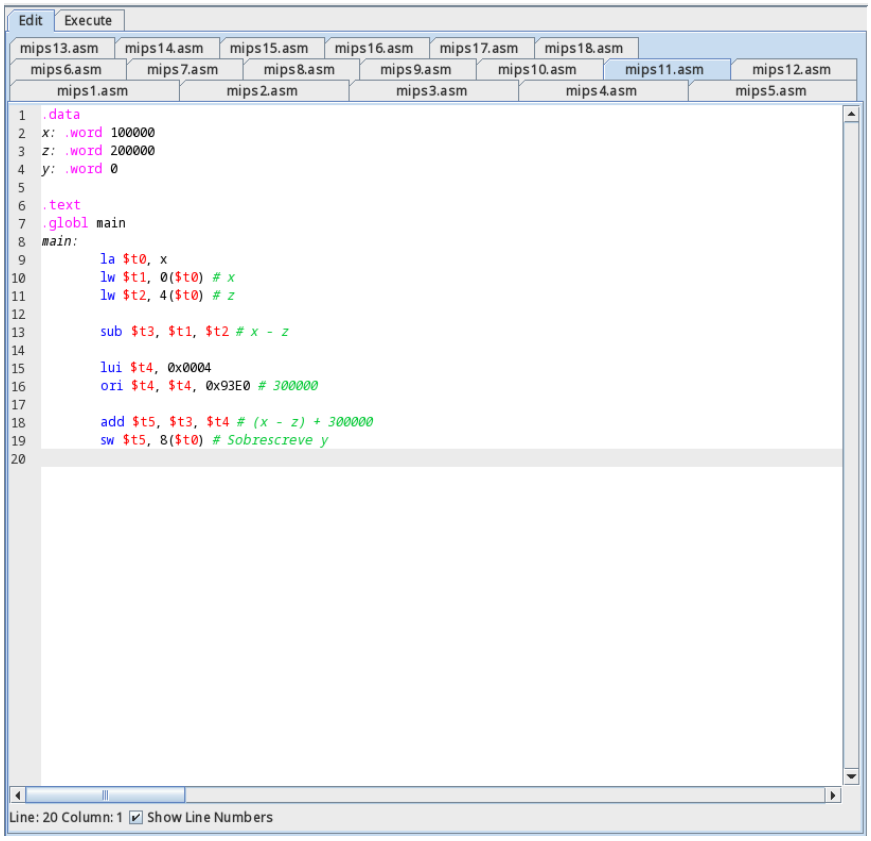

Edit
Execute
Text Segment
Bkpt Address Code Basic
0x00400000 0x3c011001 lui $1,0x00001001 11: la $t0, x1
0x00400004 0x34280000 ori $8,$1,0x00000000
0x00400008 0x8d090000 lw $9,0x00000000($8) 12: lw $t1, 0($t0) # x1
0x0040000c 0x8d0a0004 lw $10,0x00000004($8) 13: lw $t2, 4($t0) # x2
0x00400010 0x8d0b0008 lw $11,0x00000008($8) 14: lw $t3, 8($t0) # x3
0x00400014 0x8d0c000c lw $12,0x0000000c($8) 15: lw $t4, 12($t0) # x4
0x00400018 0x012a6820 add $13,$9,$10 17: add $t5, $t1, $t2
0x0040001c 0x01ab6820 add $13,$13,$11 18: add $t5, $t5, $t3
0x00400020 0x01ac6820 add $13,$13,$12 19: add $t5, $t5, $t4 # Soma
0x00400024 0xad0d0010 sw $13,0x00000010($8) 21: sw $t5, 16($t0) # Sobrescreve soma
Data Segment
Address Value (+0) Value (+4) Value (+8) Value (+c)
0x10010000 0x0000000f 0x00000019 0x0000000d 0x00000011
0x10010020 0x00000000 0x00000000 0x00000000 0x00000000
0x10010040 0x00000000 0x00000000 0x00000000 0x00000000
0x10010060 0x00000000 0x00000000 0x00000000 0x00000000
0x10010080 0x00000000 0x00000000 0x00000000 0x00000000
0x100100a0 0x00000000 0x00000000 0x00000000 0x00000000
0x100100c0 0x00000000 0x00000000 0x00000000 0x00000000
0x100100e0 0x00000000 0x00000000 0x00000000 0x00000000
0x10010100 0x00000000 0x00000000 0x00000000 0x00000000
0x10010120 0x00000000 0x00000000 0x00000000 0x00000000
0x10010140 0x00000000 0x00000000 0x00000000 0x00000000
0x10010160 0x00000000 0x00000000 0x00000000 0x00000000
0x10010180 0x00000000 0x00000000 0x00000000 0x00000000
0x100101a0 0x00000000 0x00000000 0x00000000 0x00000000
0x100101c0 0x00000000 0x00000000 0x00000000 0x00000000
0x100101e0 0x00000000 0x00000000 0x00000000 0x00000000
0x10010000 (.data)
Hexadecimal Addresses
Registers
Coproc 1
Coproc 0
Name Number Value
$zero 0 0x00000000
$at 1 0x10010000
$v0 2 0x00000000
$v1 3 0x00000000
$a0 4 0x00000000
$a1 5 0x00000000
$a2 6 0x00000000
$a3 7 0x00000000
$t0 8 0x10010000
$t1 9 0x0000000f
$t2 10 0x00000019
$t3 11 0x0000000d
$t4 12 0x00000011
$t5 13 0x00000046
$t6 14 0x00000000
$t7 15 0x00000000
$s0 16 0x00000000
$s1 17 0x00000000
$s2 18 0x00000000
$s3 19 0x00000000
$s4 20 0x00000000
$s5 21 0x00000000
$s6 22 0x00000000
$s7 23 0x00000000
$t8 24 0x00000000
$t9 25 0x00000000
$k0 26 0x00000000
$k1 27 0x00000000
$gp 28 0x10008000
$sp 29 0x7fffeffc
$fp 30 0x00000000
$ra 31 0x00000000
pc 0x00400028
hi 0x00000000
lo 0x00000000

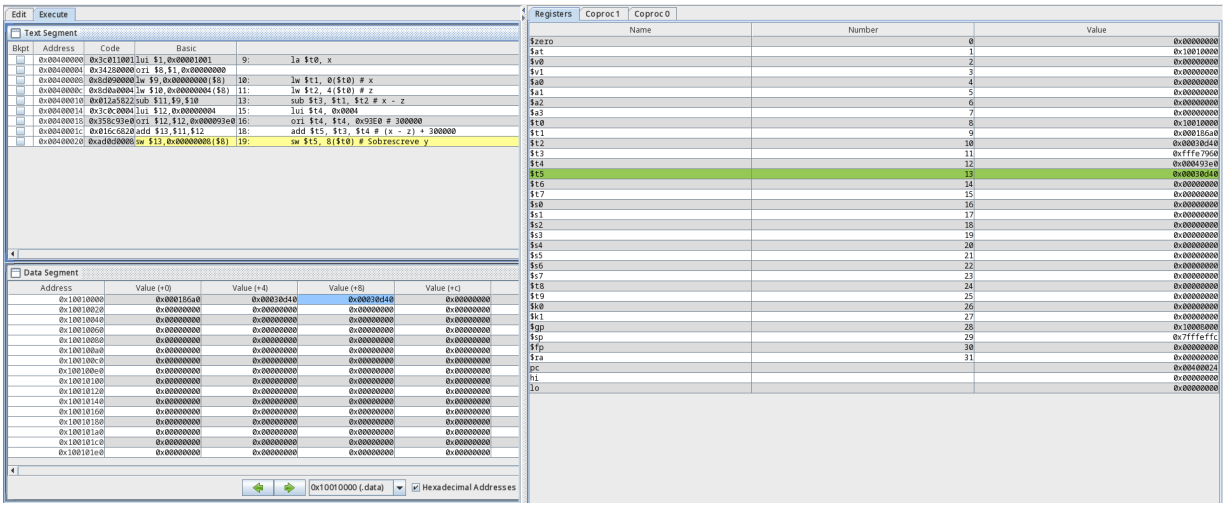
Exercício 10



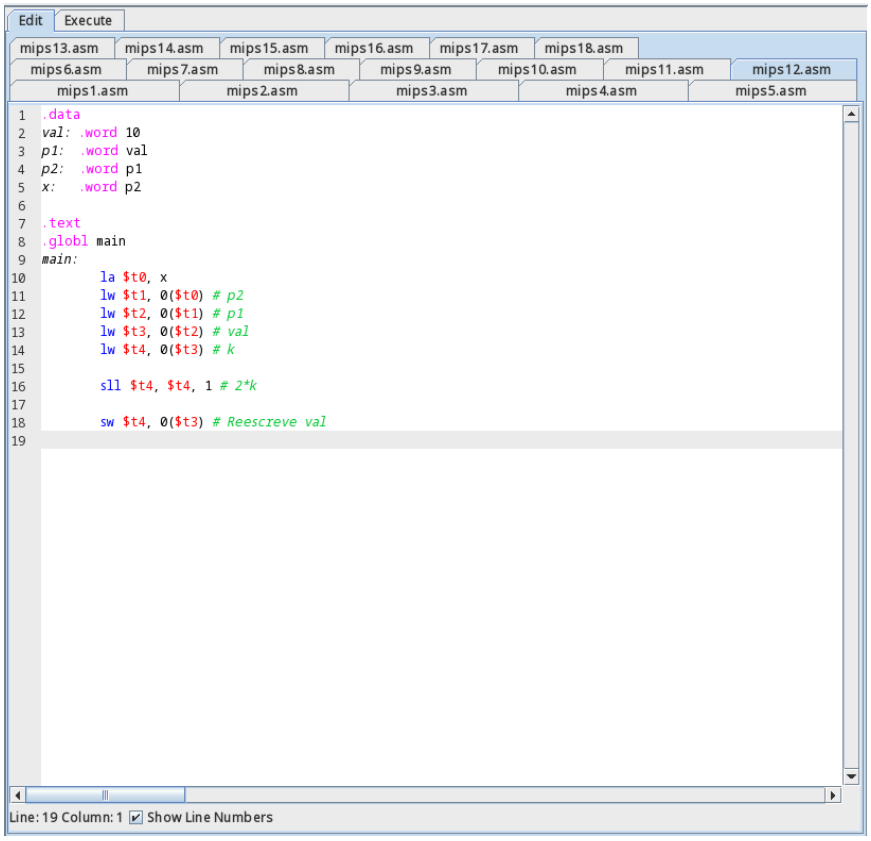


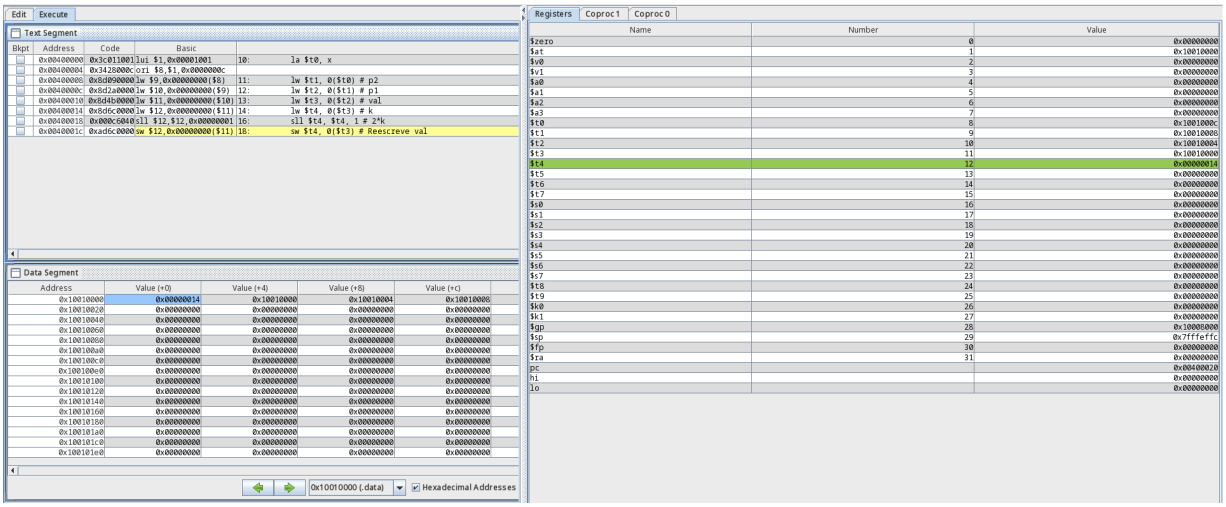
Exercício 11



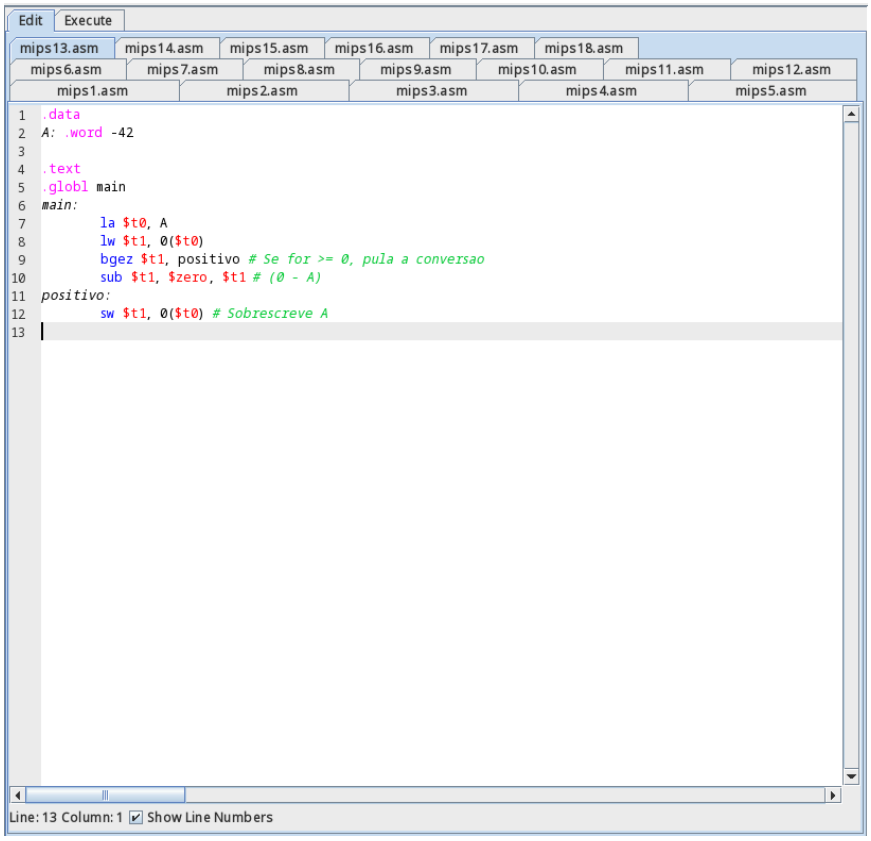


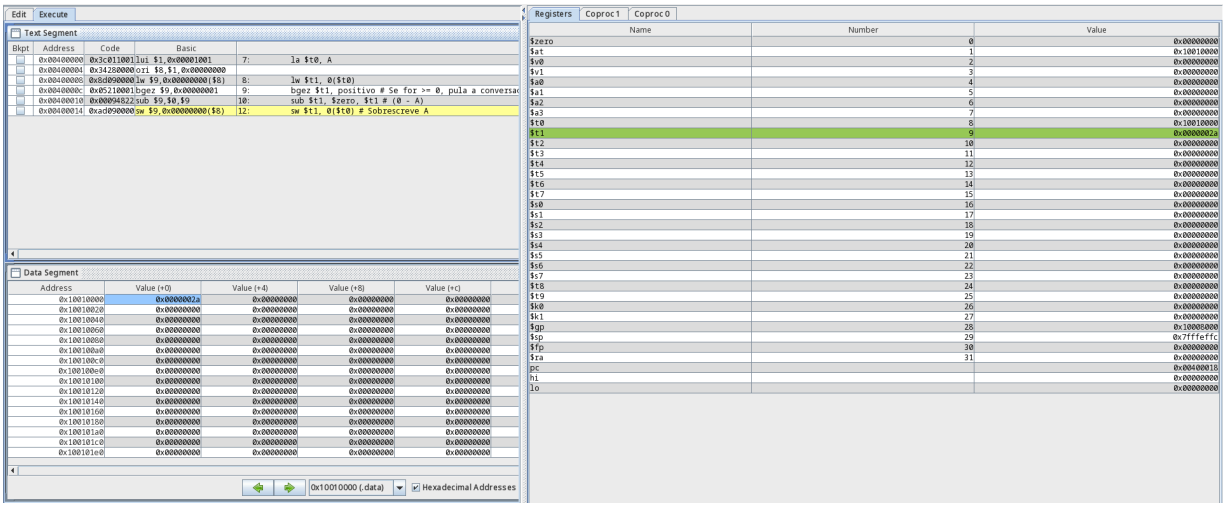
Exercício 12



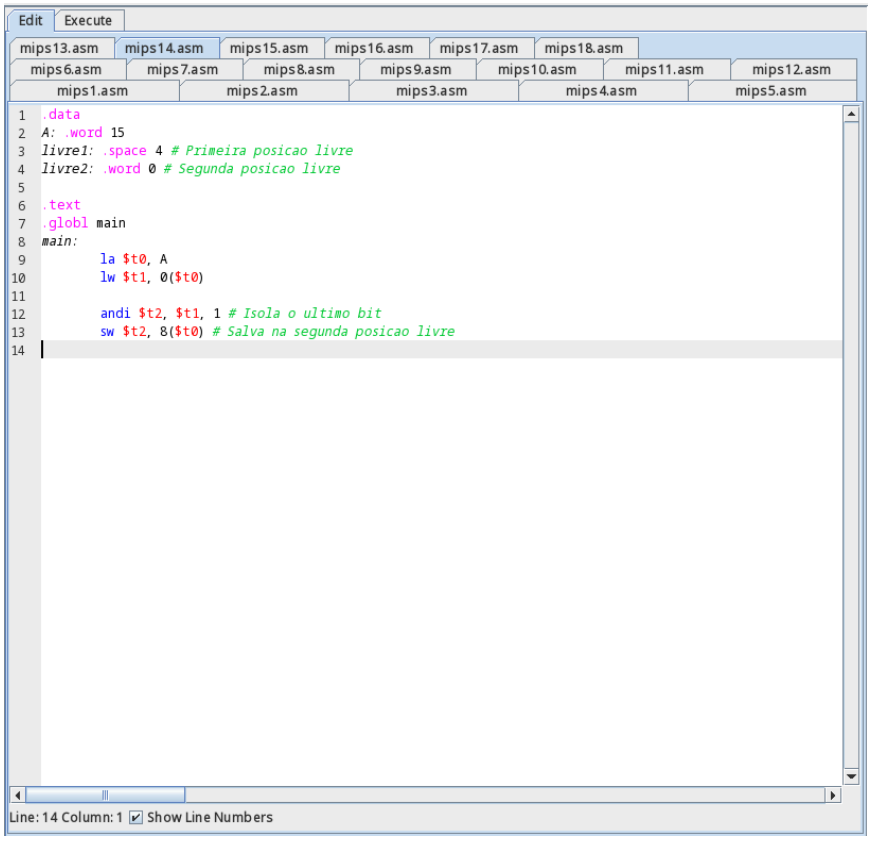


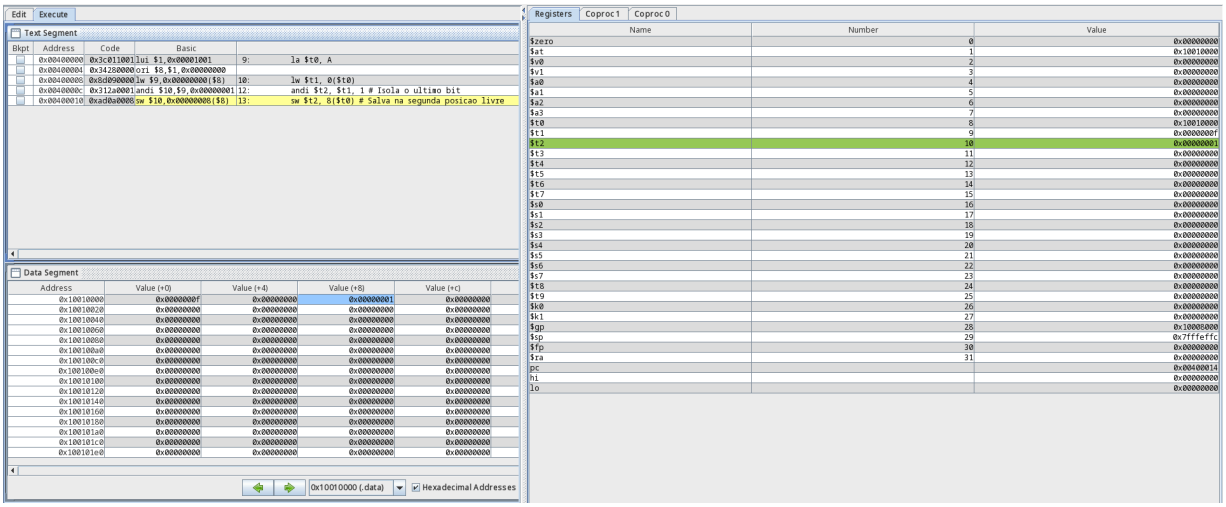
Exercício 13



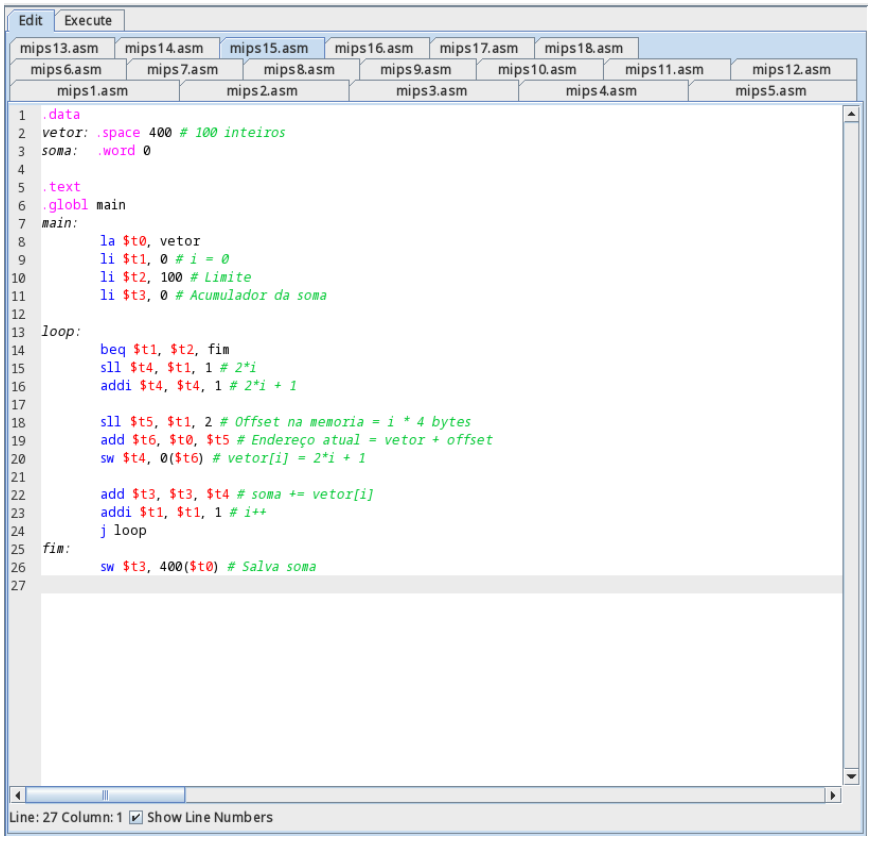


Exercício 14





Exercício 15



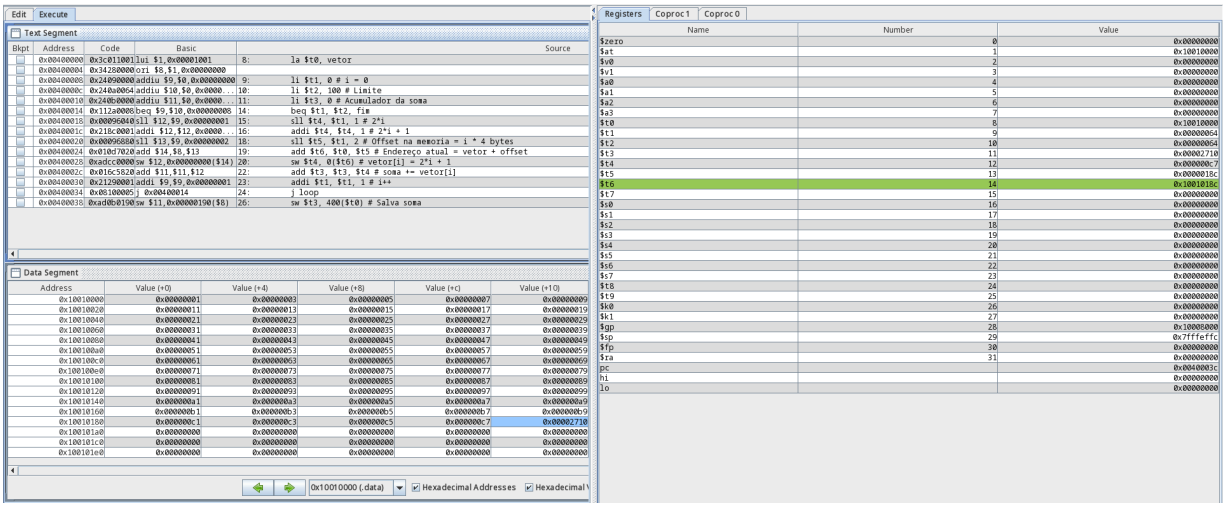
```
.data
vetor: .space 400 # 100 inteiros
soma:  .word 0

.text
.globl main
main:
        la $t0, vetor
        li $t1, 0 # i = 0
        li $t2, 100 # Limite
        li $t3, 0 # Acumulador da soma

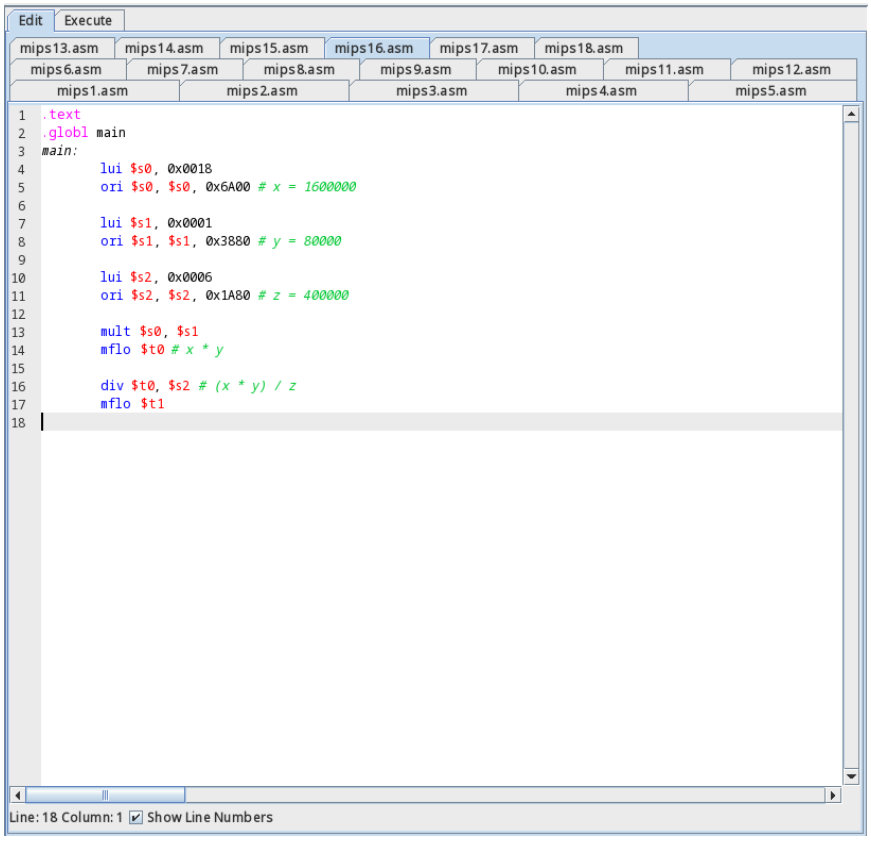
loop:
        beq $t1, $t2, fim
        sll $t4, $t1, 1 # 2*i
        addi $t4, $t4, 1 # 2*i + 1

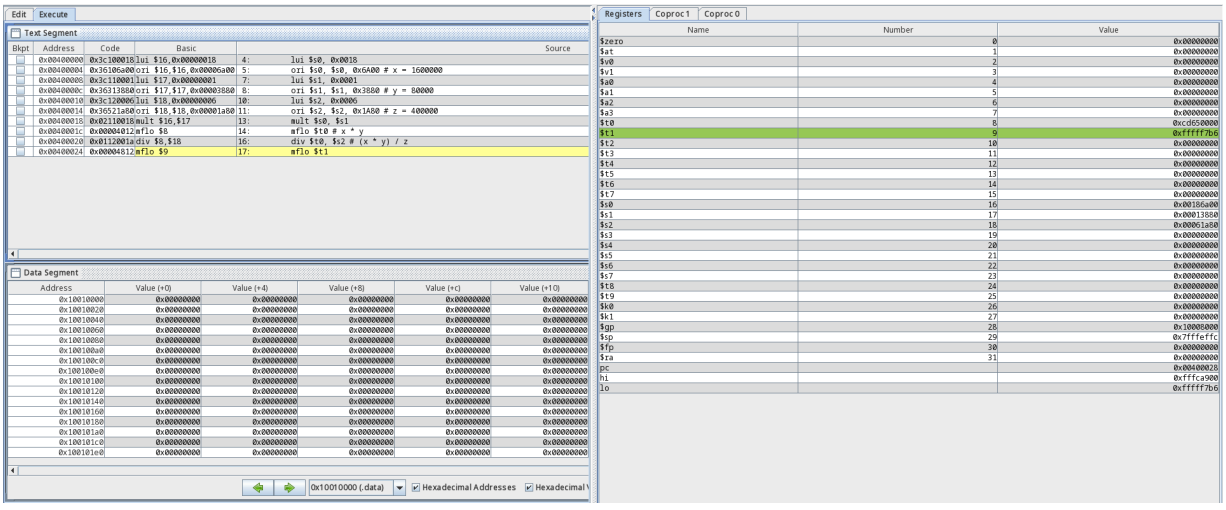
        sll $t5, $t1, 2 # Offset na memoria = i * 4 bytes
        add $t6, $t0, $t5 # Endereço atual = vetor + offset
        sw $t4, 0($t6) # vetor[i] = 2*i + 1

        add $t3, $t3, $t4 # soma += vetor[i]
        addi $t1, $t1, 1 # i++
        j loop
fim:
        sw $t3, 400($t0) # Salva soma
```



Exercício 16





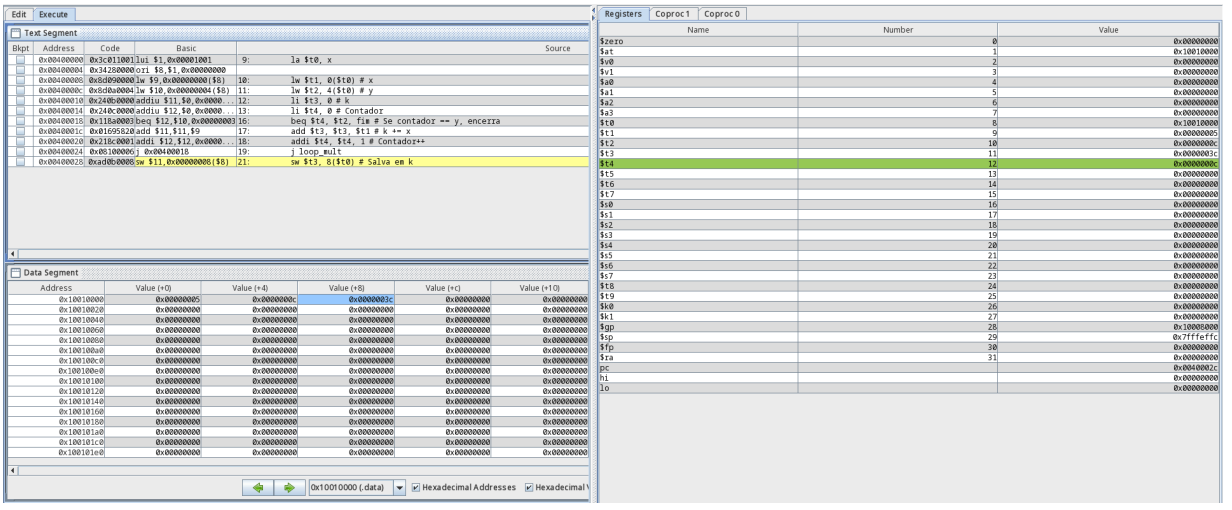
Exercício 17

```
.data
x: .word 5
y: .word 12
k: .word 0 # Terceira posicao livre

.text
.globl main
main:
        la $t0, x
        lw $t1, 0($t0) # x
        lw $t2, 4($t0) # y
        li $t3, 0 # k
        li $t4, 0 # Contador

loop_mult:
        beq $t4, $t2, fim # Se contador == y, encerra
        add $t3, $t3, $t1 # k += x
        addi $t4, $t4, 1 # Contador++
        j loop_mult
fim:
        sw $t3, 8($t0) # Salva em k
```

Line: 22 Column: 1 Show Line Numbers



Exercício 18

Edit Execute

mips13.asm mips14.asm mips15.asm mips16.asm mips17.asm mips18.asm

mips6.asm mips7.asm mips8.asm mips9.asm mips10.asm mips11.asm mips12.asm

mips1.asm mips2.asm mips3.asm mips4.asm mips5.asm

```
1  .data
2  x: .word 3
3  y: .word 4
4  k: .word 0
5
6  .text
7  .globl main
8  main:
9          la $t0, x
10         lw $t1, 0($t0) # Base x
11         lw $t2, 4($t0) # Expoente y
12
13         li $t3, 1 # k = 1
14         li $t4, 0 # i
15
16 loop_potencia:
17         beq $t4, $t2, fim_potencia
18
19         move $t5, $t3
20         li $t3, 0
21         li $t6, 0
22
23 loop_multiplica:
24         beq $t6, $t1, fim_multiplica
25         add $t3, $t3, $t5 # k += temp
26         addi $t6, $t6, 1
27         j loop_multiplica
28 fim_multiplica:
29
30         addi $t4, $t4, 1
31         j loop_potencia
32
33 fim_potencia:
34         sw $t3, 8($t0) # resultado final
35
```

Line: 35 Column: 1 Show Line Numbers

